

# Configurable Regimes: Information and Measurement

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## Why This Matters

A central goal of the simulation is to study how information structure and

accounting rules affect firm behavior, market outcomes, and financial stability. By varying these regimes across runs, we can answer questions like:

- Does more disclosure lead to better equity pricing?
- Does R&D capitalization change how much firms invest in R&D?
- Does hiding cash flow details make credit markets less efficient?
- Do firms behave differently when they know competitors can see their margins?

Both regimes are parameters of the orchestrator set in `config.yaml`. Agents do not choose what to disclose -- the orchestrator filters information before sending it.

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## Information Regime

### What It Controls

The information regime specifies, for each agent type, which fields from other agents' data are visible. It is a filter matrix: rows are observer types, columns are data fields.

### Configuration Format

```
information_regime: "baseline"
# OR specify custom:
information_regime:
  name: "custom_asymmetric"
  firms_see:
    competitor_fields: ["saleq", "niq", "atq", "ltq", "ceqq", "prccq", "cshoq"]
    competitor_ratios: ["gross_margin", "debt_equity"]
    macro: true
    environment_narrative: true
    equity_prices: true
    credit_terms_of_others: false
    competitor_rd_spend_total: true
    competitor_rd_allocation: false
    competitor_capacity: false
    competitor_unit_cost: false
  financial_institutions_see:
    firm_fields: ["saleq", "cogsq", "gpq", "xrdq", "xsgaq", "niq",
                  "cheq", "atq", "ltq", "ceqq", "dlcq", "dlttq",
                  "oancfq", "ivncfq", "fincfq", "capxq"]
    firm_ratios: ["gross_margin", "operating_margin", "debt_equity",
                  "current_ratio", "interest_coverage"]
    firm_capacity: false
    firm_unit_cost: false
```

```

firm_rd_allocation: false
firm_internal_stocks: false
other_institution_terms: false
macro: true
environment_narrative: true
environment_sees:
  all_firm_actions: true
  all_firm_private_state: true
  all_institution_portfolios: true
macro: true

```

## Preset Regimes

### 1. *baseline (default)*

The standard information structure. Reflects roughly what public markets provide.

| Observer        | Sees                               | Does Not See                       |
|-----------------|------------------------------------|------------------------------------|
| Firms           | Competitors' summary financials (r | Competitors' COGS detail, unit cos |
| Investment Bank | All firms' full published statemen | Internal stocks, R&D allocation, u |
| Commercial Bank | All firms' full published statemen | Internal stocks, R&D allocation. O |
| Credit Fund     | Same as commercial bank.           | Same as commercial bank.           |
| Environment     | Everything (god-mode).             | Nothing hidden.                    |

### 2. *full\_transparency*

Everyone sees everything. Useful as a benchmark.

| Observer   | Sees                                                |
|------------|-----------------------------------------------------|
| All agents | All firms' complete state (including internal stock |

### 3. *minimal\_disclosure*

Firms disclose almost nothing. Tests whether markets can function with scarce info.

| Observer               | Sees                                                |
|------------------------|-----------------------------------------------------|
| Firms                  | Only competitors' revenue and equity price. Macro.  |
| Financial institutions | Only revenue, net income, cash, total debt for each |
| Environment            | Everything.                                         |

### 4. *asymmetric\_banks*

Banks see more than equity investors. Tests information advantage of lenders.

| Observer        | Sees                                                |
|-----------------|-----------------------------------------------------|
| Investment Bank | Revenue, net income, total assets, equity only (lik |
| Commercial Bank | Full financial statements + cash flow + capacity ut |
| Credit Fund     | Same as commercial bank.                            |
| Firms           | Baseline.                                           |

### 5. *competitor\_intelligence*

Firms can see more about each other. Tests strategic interaction effects.

| Observer | Sees extra (vs. baseline)                          |
|----------|----------------------------------------------------|
| Firms    | Competitors' capacity, total R&D spend, capex (but |

Financial institutions

Baseline.

## How Filtering Works

The orchestrator maintains the FULL state of every agent. When constructing the context for a turn prompt, it:

- Starts with the full state
- Applies the information regime filter for the observer's type
- Removes all fields not in the allowed set
- Computes any allowed derived ratios
- Sends the filtered context to the agent

Important: The agent never knows what it cannot see. It does not receive empty fields -- the fields simply do not appear in the prompt. The agent cannot infer what is being hidden.

## How Agents Handle Missing Information

Agents are prompted to work with whatever information they have:

- "Base your decisions on the information provided. If you do not have competitor cost data, estimate it from their pricing and margins."
- "If you do not know a firm's capacity, infer it from their production volume (production cannot exceed capacity)."

This makes the simulation more realistic -- real firms must also estimate hidden competitor information.

## Measurement Regime

### What It Controls

The measurement regime specifies HOW financial statements are constructed from the underlying economic transactions. Different accounting rules produce different numbers from the same economic reality.

This affects:

- What numbers appear on the income statement, balance sheet, and cash flow statement
- How R&D, assets, and liabilities are valued
- What ratios and metrics agents compute from financial statements
- How the investment bank prices equity (different statements -> different valuations)

### Configuration Format

```
measurement_regime: "baseline_gaap"
# OR specify custom:
measurement_regime:
  name: "custom_mixed"
  rd_treatment: "expense"           # "expense" | "capitalize_partial" | "capitalize_full"
  rd_capitalization_rate: 0.0       # fraction of R&D capitalized (if capitalize_*)
  rd_amortization_quarters: 12      # amortization period for capitalized R&D
  asset_valuation: "historical"     # "historical" | "fair_value" | "lower_of_cost_or_market"
  fair_value_noise_std: 0.0         # std dev of noise added to fair value estimates
  revenue_recognition: "accrual"    # "accrual" | "cash"
  expense_recognition: "accrual"    # "accrual" | "cash"
```

```
inventory_method: "fifo"          # "fifo" | "lifo" | "weighted_average"
depreciation_method: "straight_line" # "straight_line" | "accelerated"
goodwill_treatment: "none"        # "none" | "amortize" | "impair_test"
disclosure_level: "full"          # "full" | "summary" | "minimal"
```

## Preset Regimes

### 1. *baseline\_gaap* (default)

Standard accrual accounting. R&D expensed. Historical cost.

```
rd_treatment: "expense"
asset_valuation: "historical"
revenue_recognition: "accrual"
expense_recognition: "accrual"
inventory_method: "fifo"
depreciation_method: "straight_line"
disclosure_level: "full"
```

Effect on statements:

- R&D appears as a line item on IS (xrdq), reducing operating income
- PPE carried at cost minus accumulated depreciation
- Accounts receivable and payable reflect accrual timing
- Full IS + BS + CF + SOE disclosed (subject to information regime filtering)

### 2. *rd\_capitalization*

60% of R&D spending is capitalized as an intangible asset and amortized.

```
rd_treatment: "capitalize_partial"
rd_capitalization_rate: 0.60
rd_amortization_quarters: 12
# Everything else same as baseline_gaap
```

Effect on statements:

- R&D expense on IS is only 40% of actual spending
- Remaining 60% appears as "capitalized R&D" on BS (intangible asset)
- Amortization of capitalized R&D flows through IS over 12 quarters
- Net income is higher in heavy-R&D periods (expense is deferred)
- Total assets are higher (intangible asset on BS)
- This changes equity pricing -- the investment bank sees higher earnings

and assets, potentially leading to higher valuations for R&D-heavy firms

- This may change firm behavior -- if firms know R&D is capitalized, they may invest more (reported earnings are less penalized)

### 3. *fair\_value\_assets*

PPE and inventory marked to (noisy) fair value each quarter.

```
asset_valuation: "fair_value"
fair_value_noise_std: 0.10      # 10% noise around "true" value
# Everything else same as baseline_gaap
```

Effect on statements:

- PPE and inventory revalued each quarter
- Gains/losses flow through other comprehensive income (OCI) or IS
- Balance sheet is more volatile but potentially more informative
- Noise introduces uncertainty -- fair value is estimated, not exact
- Investment bank must decide how much weight to give noisy fair values

#### 4. *cash\_basis*

No accruals. Revenue recognized when cash received. Expenses when cash paid.

```
revenue_recognition: "cash"
expense_recognition: "cash"
# No AR, no AP, no accrued expenses
```

Effect on statements:

- No accounts receivable or payable on balance sheet
- Revenue = cash collections (lagged vs. accrual)
- Expenses = cash payments
- IS is more volatile (timing of cash flows != timing of economic activity)
- Simpler balance sheet but potentially misleading about economic reality

#### 5. *minimal\_disclosure*

Firms report only summary numbers. Fewer line items on statements.

```
disclosure_level: "minimal"
# Income statement: only revenue, operating_income, net_income
# Balance sheet: only total_assets, total_liabilities, total_equity, cash
# Cash flow: only total_change_in_cash
# No component breakdowns
```

Effect: Financial institutions and competitors have less data to analyze. Credit underwriting becomes harder. Equity pricing becomes noisier.

#### 6. *expanded\_disclosure*

Firms report additional operational metrics beyond financial statements.

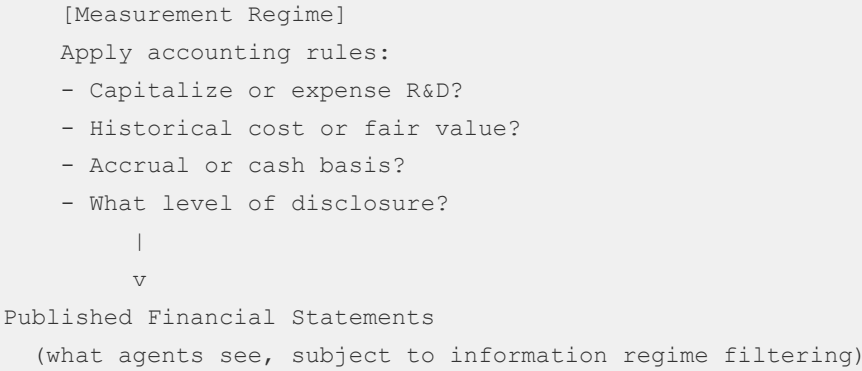
```
disclosure_level: "expanded"
# Everything in baseline_gaap, PLUS:
# - Capacity utilization (%)
# - R&D spending by program (product/process/delivery)
# - Product generation
# - Backlog / unfilled orders
# - Employee count (proxy for scale)
```

Effect: More information available to all observers. May improve equity pricing accuracy and credit underwriting.

### How Measurement Regimes Are Applied

The orchestrator applies the measurement regime when computing financial statements from the underlying economic data:

```
Underlying Economic Reality
(transactions, stocks, flows -- maintained by orchestrator)
|
v
```



The underlying economic reality is always the same. Only the REPORTING differs. This means:

- The orchestrator tracks both the "true" economic state AND the reported state
- Feasibility clamping and solvency checks use the TRUE state (real cash, real debt)
- Agent decisions are based on the REPORTED state (what they see on statements)
- Equity pricing is based on REPORTED financials
- But defaults are triggered by REAL cash positions

This creates a natural tension: if the measurement regime makes firms look healthier than they are (e.g., R&D capitalization inflates assets), the market may misprice equity and extend too much credit -- until reality catches up.

## Combining Information and Measurement Regimes

The two regimes are orthogonal and can be combined freely:

| Combination                            | What It Tests                                       |
|----------------------------------------|-----------------------------------------------------|
| baseline info + baseline GAAP          | Standard modern capital market                      |
| baseline info + R&D capitalization     | Does capitalizing R&D improve investment efficiency |
| minimal info + baseline GAAP           | Can markets function with limited disclosure?       |
| full transparency + baseline GAAP      | Does more info improve outcomes? (benchmark)        |
| baseline info + minimal disclosure     | Double opacity -- limited fields AND limited line i |
| asymmetric banks + expanded disclosure | Banks as informed lenders -- does it reduce default |

## Experimental Design

A typical experimental run plan:

```
experiments:
- name: "baseline"
  information_regime: "baseline"
  measurement_regime: "baseline_gaap"
  n_runs: 5
  seeds: [1, 2, 3, 4, 5]

- name: "rd_capitalization_effect"
  information_regime: "baseline"
  measurement_regime: "rd_capitalization"
  n_runs: 5
  seeds: [1, 2, 3, 4, 5]    # same seeds for comparability

- name: "transparency_effect"
  information_regime: "full_transparency"
```

```
measurement_regime: "baseline_gaap"  
n_runs: 5  
seeds: [1, 2, 3, 4, 5]
```

By holding seeds constant across experiments, the exogenous shocks are identical, and differences in outcomes are attributable to the regime change.

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## Implementation Notes

### Where Regime Logic Lives

- Information regime: Applied in the orchestrator's prompt builder.

A `filter_context(full_state, observer_type, info_regime)` function strips fields before sending to agents.

- Measurement regime: Applied in the orchestrator's accounting module.

A `build_statements(economic_state, measurement_regime)` function produces the published financial statements.

### Adding New Regimes

To add a new regime:

- Define it in `config/regimes/` as a YAML file
- Implement any new accounting logic in `accounting/postings.py`
- The filter and statement builder read the regime config dynamically
- No changes needed to agent code -- agents just see different data

### Regime Is Fixed Within a Run

The information and measurement regimes are set at run start and do not change during the run. (A future extension could model regime changes -- e.g., a regulatory mandate to capitalize R&D mid-simulation.)